Silicon & Handset — Master Counsel, Execution Playbook & Multi-AI Engineering Charter

Addressed Directly To: `GPT Sol 6.1` & `Gemini 4 Argon` (Primary Implementing AI Engineers)
Written On Behalf Of: The Project Owner (Game Visionary & Non-Programmer Observer)
Target Repository / Branch: `spinoff-phone` (`SPINOFF_SOURCE/`)
Companion Specifications in This Folder:

1. `AGENTS.md` & `BRANCH_LAYOUT.md` — Immutable constitutional rules and path mappings.
2. `ACTIVE_PROMPT.md` — `v0.3.0` Component SKU/Lot provenance & firm-specific R&D specification.
3. `DEEP_LOGIC_BUGS_AND_FUTURE_TIMEBOMBS_AUDIT` — 10 active code bugs & latent scalability/clock defects to fix first.
4. `ECONOMIC_AND_BUSINESS_REALISM_AUDIT_REPORT` — Macroeconomic velocity closure, Multinomial Logit demand, scale calibration & insolvency fixes.
5. `SUPPLIER_ECOSYSTEM_AND_SEED_DIVERGENCE_BLUEPRINT` — Meso-scale component physics & heavy-tailed procedural NPC/world genesis.
6. `LIVING_ECOSYSTEM_ENGINE_AND_AUTONOMOUS_DIFF_SPEC` — Combinatorial BOM solver, 8-axis physical benchmarks, hedonic pricing & 8 stateful consumer personas.
7. `POST_V030_ARCHITECTURE_AND_ROADMAP_HANDOFF` — Phased post-`v0.3.0` roadmap (`v0.4.0` -> `v0.6.0`: installed base, RMA, OTA, foundry queues, escrow, boardroom dissent, channels/leaks).

---

1. Understand Your Human Owner First (Critical Context for Both AI Models)

The owner of this project **does not write code, does not debug JavaScript consoles, and is not a smartphone hardware engineer**. They are building this game because they love **watching a living, breathing, causally honest miniature world**—where companies, engineers, suppliers, and customers make real decisions, triumph, blunder, adapt, and surprise the observer on every new seed.

This single fact changes your responsibility as an AI engineer:

1. **Never hand back half-wired plumbing, broken standalone HTML builds, or raw technical jargon in the UI.** Every patch you deliver must open cleanly offline via

`Silicon_Handset_single.html` on both mobile (`390px`) and desktop (`1280px`) with zero external network/CDN dependencies.

2. **Do not wait to be micromanaged on formulas or edge cases.** Exercise proactive architectural judgment within `AGENTS.md` invariants, verify your own work ruthlessly with automated tests and multi-year soaks, and leave crystal-clear handoff notes so the *other* AI (`GPT Sol 6.1` $\leftrightarrow$ `Gemini 4 Argon`) can continue seamlessly on the next ZIP turn.

---

2. Avoid "Invisible Math Syndrome": Turn Deep Physics into Watchable Stories

The #1 failure mode of AI-built simulation games is writing brilliant backend math in `sim.js` while rendering an unreadable spreadsheet of 50 raw numbers in `ui.js`. **If a non-technical observer cannot tell *what* happened and *why* within 5 seconds, your simulation depth is invisible.**

2.1 Mandatory Observability & Game-Feel Upgrades

1. **Causal "Why" Chains Over Raw Numbers:** Every major NPC action—choosing a chip supplier, canceling an R&D project, slashing price in clearance, rejecting a job offer, or losing market share—must store a structured causal record rendered into natural, specific Thai:
   - *Bad (Generic):* `"กำลังวิจัยและประเมินตลาด"`
   - *Good (Causal & Specific):* `"เลือกใช้ชิป B-Bin ของ อิสระเซมิคอนดักเตอร์ คู่กับตัวถังอะลูมิเนียม เพราะชิปตัวท็อปกินไฟเกินพิกัดแบตเตอรี่ที่มีในตลาด และงบต่อเครื่องกลุ่มคุ้มค่าจำกัดที่ ฿2,900"`
2. **Side-by-Side Comparison & Bottleneck Badges:**
   - Add a **2–3 Phone Side-by-Side Comparison Drawer** in `รุ่นและสเปก` highlighting winners in green and physical bottlenecks in amber/red (e.g., 🔴 `ร้อนจัด 47.5°C เฟรมเรตตก 32%`, 🟢 `กล้องกลางคืนอันดับ 1 ของโลกนี้`).
   - Add a **"World Industry Headline / Turning Points" Digest** summarizing major structural shifts (breakthrough SKUs, market-leader dethronements, star engineer poaching, supplier stockouts, liquidity rescues).
3. **Expressive Live Campus Canvas (`drawCampus`):** Keep rendering lightweight 2D Canvas, but make it *informative and alive*: show distinct lab zones (Design/SoC, Optics/Camera, SMT/Pilot Line, Commercial/Boardroom, Off-Duty), subtle status indicators above NPCs (e.g., inspired breakthrough, thermal-bug firefighting, exhausted/high stress, serving notice to leave), and click-to-inspect NPC cards!
4. **1-Click "What-If" Scenario Presets in `Edit Hub`:** Alongside manual numeric inputs for cash, subjects, and traits, give the owner fun, logged **Experimental Scenario Buttons** (still 100% compliant with `AGENTS.md` experiment tracking):

- *Inject a Generational Specialist* into a struggling underdog firm.
- *Trigger a 14-Day Upstream Silicon/Glass Supply Disruption* and watch brands scramble for alternative SKUs.
- *Clone Snapshot Baseline* before an edit so the UI displays **"ผลต่างเทียบกับก่อนทดลอง (Delta vs. Pre-Edit Baseline)"** after 30/90 days!

---

3. Master Execution Order (How to Sequence All 6 Blueprints Without Breaking the Game)

Do **not** attempt to rewrite everything in a single untested 4,000-line commit. Follow this exact dependency-safe sequence:

Phase 1 — Foundation Hardening & Logic Bug Exorcism (Do First!)

Apply `DEEP_LOGIC_BUGS_AND_FUTURE_TIMEBOMBS_AUDIT`:

- Fix **`LB-01`** (`recruit()` iterating across all willing/affordable candidates instead of aborting on `candidates[0]`).
- Fix **`LB-02`** (include `displayQualityScore` in `consumerUtility` and fix GPU fill-rate gaming math so better displays aren't net-negative poison).
- Fix **`LB-03`** (monotonic hour-boundary timestamps in `advance()` so mid-hour `Edit Hub` commands never precede `minute: 0` ticks).
- Fix **`LB-04`** **&** **`LB-05`** (map all 7+ subjects to multi-aptitude `HumanCore` specs instead of `potential.memory`, wire `H.evaluateAbility()`/`H.workplaceVoice()`, and use `'offDuty'` for evening hours to prevent `sleepDebt` burnout spirals).
- Fix **`LB-09`** **&** **`LB-10`** (preserve DOM focus/`<select>` state during `1x/5x/20x` playback; separate personal meal micro-transfers from corporate financial history so `w.ledger` doesn't wipe in 3.5 days; remove hardcoded `quantity * 7` / `batch === 12` magic constants from `audit()`).

Phase 2 — True Component Provenance, Meso-Scale Suppliers & Seed Divergence (`v0.3.0` Core + Blueprint 5)

Implement `ACTIVE_PROMPT.md` + `SUPPLIER_ECOSYSTEM_AND_SEED_DIVERGENCE_BLUEPRINT`:

- Split `dram_memory` and `nand_storage` into independent supplier roles alongside `chip`, `camera_module`, `display_panel`, `battery_pack`, `enclosure`, and `ems`.
- Implement heavy-tailed NPC genesis (Apprentices, Journeymen, Seniors, and rare `HumanCore`-backed Domain Luminaries), asymmetric firm founding archetypes, and 5

differentiated upstream material feeds so every New World seed develops a distinct technological superpower and bottleneck.
- Wire supplier Corporate DNA, RFQ win/loss R&D steering, D_0 wafer/panel yield learning, `A-Bin`/`B-Bin` lot splitting, and dynamic inventory-cover pricing.

Phase 3 — Combinatorial Handset Architecture, 8-Axis Benchmarks & Living Market (`v0.3.x` / `v0.4.0` + Blueprints 2 & 3)

Implement `LIVING_ECOSYSTEM_ENGINE_AND_AUTONOMOUS_DIFF_SPEC` + `ECONOMIC_AND_BUSINESS_REALISM_AUDIT_REPORT`:

- Replace `candidates[0]` winner-take-all demand with **Seeded Multinomial Logit + Outside Option** across **8 Stateful Consumer Archetypes**.
- Close the **Stock-Flow Consistent Circular Cash Loop** (recycling employee living spend and upstream factor income back into consumer cohorts while keeping `totalCash(w) === openingCapital + editCapital` exact to the satang).
- Calibrate wages, segment ASP ceilings, MOQ/batch setup costs, and custom R&D NRE to the simulated market scale ($Q_{\text{breakeven}} \approx 10%\text{–}25%$ of annual addressable volume).
- Implement 4-layer **Hedonic Equilibrium & Sunk-Cost Clearance Pricing** plus **Insolvency / Distress Asset Liquidation** to eliminate immortal zombie firms.

Phase 4 — Durable Installed Base, Post-Sale RMA/OTA, Escrow & Boardroom Drama (`v0.4.0` -> `v0.6.0` + Blueprint 1)

Implement `POST_V030_ARCHITECTURE_AND_ROADMAP_HANDOFF`:

- Add weekly vintage `installedBase` buckets, thermal/battery degradation, warranty `RMA` queues with root-cause engineering lessons, `brandTrust`, and post-launch `OTA` firmware updates.
- Add staged `escrow` purchase commitments (pre-launch Working-Capital Valley), multi-hat C-suite `boardProposals` with inspectable dissent, intra-lab mentoring, and B2B channel `sell-in`/`sell-through` + leak/review dynamics.

---

4. Pacing & Numerical Calibration Rules ("Feel Good at 1x, Survive 3,000 Days at 20x")

Calibrate time constants and thresholds against these concrete gameplay benchmarks:

1. **First 30 Simulated Days:** Suppliers quote and ship opening lots; brands assemble and launch initial entry-level models; first competitive price/share splits appear; observer sees distinct NPC schedules and initial R&D project choices.
2. **Days 60–180:** Divergent R&D projects complete (and 15%–30% stumble into rework/respins!); first talent poaching transfers occur; `brandTrust` and supplier D_0 yield gaps begin separating smart specialists from overextended gamblers.
3. **Years 1–3 (`Days 365–1,095`):** Multi-generation transitions occur organically; household circular cash velocity sustains healthy annual demand (`3,500–12,000` units/year); poorly cooled phones suffer visible RMA warranty waves; insolvent firms restructure or liquidate cleanly without crashing or freezing the economy.
4. **Performance Budget:**
    - Node simulation tick (`60` simulated minutes): **`< 1.5 ms` mean** even in Year 3 (`1,095` days).
    - Browser standalone file size (`Silicon_Handset_single.html`): keep clean, zero bloat, 100% offline-playable (`file://` and `http://` safe—never use `fetch()`, ES module `import` over `file://`, WebWorkers requiring external URLs, or remote fonts).

---

5. Multi-AI Collaboration Protocol (`GPT Sol 6.1` $\leftrightarrow$ `Gemini 4 Argon`)

Because two different AI models will alternate working on this codebase via owner-transferred ZIP files:

1. **Single Canonical Module Manifest:** If you split `SPINOFF_SOURCE/js/` into modular UMD files (`component-physics.js`, `supplier-ai.js`, `brand-architect-ai.js`, `living-market.js`, `sim.js`, `ui.js`), define their exact load order in **one shared manifest** (`SPINOFF_SOURCE/tools/manifest.cjs` or `package.json` metadata) consumed identically by `index.html`, `tools/build.cjs`, `tools/check.cjs`, and `test/*.cjs`. Verify `Silicon_Handset_single.html` === `SPINOFF_SOURCE/Silicon_Handset_single.html` byte-for-byte after `npm run build`!
2. **Never Leave Ghost Code or Half-Migrated Schemas:** When bumping `w.schema` (e.g., `2 -> 3 -> 4`) and `localStorage` key (`silicon-handset-v030`, `v040`), update `audit(w)` and `deserialize()` atomically in the same patch. Explicitly reject older schema saves with a polite Thai message without deleting old `localStorage` keys or overwriting corrupt autosaves.
3. **Strict Integer & PRNG Discipline:**
    - Money: safe non-negative integers in satang (`Number.isSafeInteger`).

    - Work & aging accumulators: prefer integer milli-units (`1000 = 1.0`) over raw floats to prevent cross-engine float drift.
    - PRNG: pure deterministic hash/PRNG keyed by `(world.seed, domain, entityId, tickCounter)`—never call `Math.random()` or `Date.now()` inside simulation logic!
4. **Mandatory Handoff Breadcrumbs Every Turn:** Before packaging your output ZIP for the owner, always update:
    - `START_HERE_FOR_AI.md` & `project_status_and_ai_onboarding.md`
    - `spinoff_phone/PROJECT_STATUS.md` (separating **Implemented & Tested** vs. **Pending Next Patch** honestly)
    - `spinoff_phone/WORK_STATUS.md` & `PATCH_REPORT_V0.X.0.md` (recording received base SHA, changed files, test/soak results, and exact recommended next step for the next AI).

---

6. Final Golden Rule

Respect reality, respect conservation, and respect the observer's curiosity.
Never fake an outcome with a random text generator or a hidden subsidy. When a tiny 3-person Thai camera-module boutique hires an overlooked optics genius, perfects a low-light OIS module, wins contracts from three major phone brands, and out-earns a reckless SoC giant that burned its cash on a failed 28nm tape-out—**and every single satang, component lot, watt of heat, and engineer's career step in that story is 100% real and inspectable**—you will have built the exact game the owner dreamed of.